FIELD COMPARISON OF PM_{10} SAMPLERS

L. Guo, R. G. Maghirang, E. B. Razote, J. G. Tallada, J. P. Harner III, W. L. Hargrove

ABSTRACT. *Fugitive emissions of PM_{10} (particulate matter with equivalent aerodynamic diameter of 10 μm or less) are an increasing concern for concentrated animal feeding operations, including open-lot beef cattle feedlots. Various federal reference method (FRM) and equivalent samplers can be used to measure PM_{10} concentrations. This research compared the performance of Tapered Element Oscillating Microbalance ™ (TEOM), FRM high-volume (HV), and low-volume (LV) PM_{10} samplers in measuring PM_{10} concentrations in the vicinity of large cattle feedlots in Kansas. Each of the three samplers was installed at the downwind and upwind perimeters of two commercial cattle feedlots (KS1 and KS2). Samplers were operated from May 2006 to February 2008 at KS1 and from February 2007 to February 2008 at KS2. PM_{10} concentration ranged from <10 to 832 μg m^{-3} at KS1 and from <10 to 713 μg m^{-3} at KS2. Comparison of collocated PM_{10} samplers showed that measured PM_{10} concentration was generally largest with the TEOM PM_{10} sampler and smallest with the LV PM_{10} sampler. Differences in PM_{10} concentration among samplers were affected by location and duration of sampling, season, and slightly by weather conditions.*

Keywords. *PM_{10} mass concentration, High-volume sampler, Low-volume sampler, TEOM, Cattle feedlot.*

Fugitive emissions of particulate matter (PM), ammonia, and odor are an increasing concern for animal feeding operations (AFOs), including open-lot beef cattle feedlots and dairies. Dust generated from cattle feedlots has potential to cause a number of respiratory problems in humans as well as livestock (Purdy et al., 2004). Inhalation of fine PM is believed to be the major cause of increased health risks (Seaton et al., 1995). As a result, PM_{10} (PM with equivalent aerodynamic diameter of 10 μm or less) and $PM_{2.5}$ (PM with equivalent aerodynamic diameter of 2.5 μm or less) are being regulated (Purdy et al., 2007). For example, primary and secondary 24-h national ambient air quality standards for PM_{10} are 150 μg m^{-3} and not to be exceeded more than once per year on average over 3 years (Chow et al., 2006).

Measuring PM_{10} in cattle feedlots is necessary to investigate health effects posed by PM_{10} emissions, monitor transport and fate of PM_{10}, and assess effectiveness of PM abatement measures. The Federal Reference Method (FRM) and Federal Equivalent Method (FEM) of measuring PM emissions are recommended by the U.S. EPA (U.S. EPA, 1987). The FRM is a gravimetric technique in which particles are collected on a filter that is exposed for 24 h (Bulpitt and Price, 2006). The equivalent PM measurement methods, including the Tapered Element Oscillating Microbalance (TEOM), have been used to provide direct PM mass measurement for near real-time continuous data acquisition (Meyer et al., 2000; U.S. EPA, 2004).

Measurement technologies for PM are affected by the complexity of PM emissions. Actual constituents and particle size vary with geographical location and meteorological conditions (National Research Council, 2003). Inorganic, semi-volatile or volatile components, sulfured hydrogen, and ammonium nitrate may be present in the particulate phase and may be lost with time and location (Allen et al., 1997; Jerez et al., 2006). Currently, effectiveness of the measurement method is one of the more important issues in PM quantification and control, and analytical methods, including consideration of sampling uncertainties, should be fully understood (Chow, 1995; Bulpitt and Price, 2006). Several instruments are currently being used to measure PM_{10} concentration. Comparison of these samplers is needed to determine agreement of these samplers and improve various designs that influence what is actually being measured.

The major objective of this study was to compare PM_{10} concentrations measured by three different PM_{10} samplers [i.e., low-volume (LV), FRM high-volume (HV), FEM TEOM] under field conditions. Performance of similar PM_{10} samplers placed in two commercial cattle feedlots in Kansas was also evaluated. Results may provide an understanding of the PM_{10} measurement techniques as well as improve PM_{10} measurement in future studies.

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MATERIALS AND METHODS

DESCRIPTION OF PM_{10} SAMPLERS

This study considered the FRM HV PM_{10} sampler (Thermo Fisher Scientific, Franklin, Mass.), LV PM_{10} sampler (2100 Mini-Partisol, Thermo Fisher Scientific, Franklin, Mass.), and TEOM PM_{10} sampler (Series 1400a, Thermo Fisher Scientific, Franklin, Mass.; EPA designation

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No. EQPM-1090-079). The LV and HV PM_{10} samplers with PM_{10} size-selective inlets are gravimetric measurements that yield time-integrated mass concentration of PM_{10}. During measurement, ambient air is drawn into the size-selective inlet of the sampler using a vacuum pump. As ambient air passes through a filter, PM is collected on the filter. Mass of the collected PM is calculated by subtracting the gross weight of the sample filter from its tare weight. Mass of PM_{10} is then divided by the sampling flow volume to get the mass concentration of PM_{10}.

Flow rate is critical for particle fractionation and calculation of mass concentration. For the HV sampler, the flow rate of 1130 L min^{-1} is controlled by venturi (Thermo Electron Corp., 2003). For the LV sampler, the flow control system uses a temperature and pressure compensated mass flow control scheme to maintain a constant volumetric flow rate of 5 L min^{-1} (Rupprecht & Patashnick Co., Inc, 2004a).

The TEOM PM_{10} sampler, a FEM sampler for PM_{10} measurement, is also a filter-based mass measurement technique. An inertial balance directly measures the mass collected on the filter cartridge every 2 s by monitoring the corresponding frequency changes of a tapered element. The sample flow passes through the filter and then continues through the hollow tapered element on its way to an active volumetric flow control system and vacuum pump. The total air flow rate of 16.67 L min^{-1} is separated into a filter flow rate of 3 L min^{-1} and auxiliary flow rate of 13.67 L min^{-1}. The airstream for the filter flow is heated to 50°C to remove particle-bound water (Rupprecht & Patashnick Co., Inc., 2004b; Jerez et al., 2006). The TEOM system has near real-time data output and near real-time mass measurement capability. Furthermore, it eliminates possible filter handling errors that might happen with manual and gravimetric methods (Green and Fuller, 2006).

Site Description

Measurements were conducted at the upwind and downwind perimeters of two cattle feedlots, herein referred to as KS1 and KS2, in Kansas. KS1 is located approximately 40 km southwest of KS2. Prevailing wind directions at the sites were south-southeast in summer and north-northwest in winter. Annual average precipitation in KS1 and KS2 were approximately 573 and 671 mm, respectively. Cattle capacities of KS1 and KS2 are 30,000 and 25,000 head, respectively. For both feedlots, feed was processed and mixed in the feedmill, loaded on feed trucks, and delivered to the pens three times a day. The major source of PM emission for both feedlots was the pen surface. As the cattle moves around the pen, the accumulated manure on the pen surface is ground into smaller sizes that can be subsequently suspended in the air. This is generally more pronounced during the late afternoon to early evening hours when the cattle are most active. For PM control, KS1 has a solid set sprinkler system that was normally operated from May to October and during times when there was a prolonged dry period. In addition, pens at KS1 were scraped year round and manure was harvested at least twice a year. At KS2, pens were also scraped year-round and manure was harvested five to six times a year. The soil at KS2 was generally clay loam, resulting in better compaction of the base soil and better scraping of the pens compared to KS1, which had sandy soil.

A set of TEOM PM_{10}, HV PM_{10}, and LV PM_{10} samplers was set up along the north and south perimeters of KS1 in April 2006 (fig. 1). Similarly, TEOM PM_{10} and HV PM_{10} samplers were set up in February 2007 and LV PM_{10} samplers were set up in September 2007 at the north and south perimeters of KS2. To compare similar samplers, two identical HV PM_{10} and LV PM_{10} samplers were used from April to December 2006 and from July and August 2007, respectively, at both of the north and south sampling locations of KS1. In addition, there were two identical TEOM PM_{10} samplers at the south sampling location of KS1 from January to July 2007.

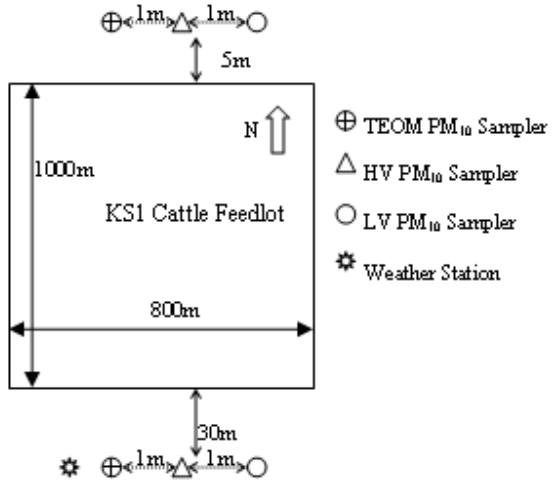


Figure 1. Schematic diagram showing the location of the PM_{10} samplers and the weather station at KS1 (not drawn to scale).

A weather station was installed at the south sampling location of KS1 and north sampling location of KS2 to obtain meteorological information. Weather data from a local weather station were also collected.

Air Sampling

At each sampling location, samplers were placed side by side with a minimum distance of about 1 m from each other (fig. 1). Intensive sampling campaigns (approximately 24 h) were conducted monthly from May 2006 to February 2008, and six 5-day intensive sampling events were conducted for the months of July to October 2007 and February 2008. For each run, the sampling duration was from 2 to 24 h (107 runs with duration of 6 h or greater; 13 runs had a duration less than 6 h at KS1; 57 runs had a duration of at least 6 h at KS2). The sampling duration was initially from 2 to 4 h and was increased to 6 to 24 h, particularly during events of low concentration, to increase the mass of PM collected on the filter

Filters used for HV PM_{10} samplers were 20- × 25-cm, type A/E glass fiber filters (Gelman Sciences, Ann Arbor, Mich.), and either Pallflex TX40 or PTFE filters (Whatman Inc., Clifton, N.J.) were used for the LV PM_{10} samplers. All filters were conditioned in the conditioning chamber [25°C, 40% relative humidity (RH)] for 24 h before weighing, before and after sampling, to minimize the effect of humidity. Field blank filters were used for the HV and LV samplers for quality control of the data. Blank filters were handled in the same way as the other filters, the only difference was that blank filters were placed in the samplers and then taken out without turning on the pump (i.e., no sampled air going through the filter). Blank filters were used in 79% and 62%

of the total runs for the HV and LV samplers, respectively. The average difference in the initial and final weight of the blank filters was approximately 1 mg for the HV samplers and 0.03 mg for the LV samplers. On the average, the mean values of the change in mass of the field blanks as percentage of the particulate mass on the collection filters were 7% (6% for the north or downwind location and 8% for the south or upwind location) and 26% (21% for the north location and 31% for the south location) for the HV and LV samplers respectively. In accordance with U.S. EPA's recommendation (U.S. EPA, 1998), the field blanks were not used for correction but as a quality control check to detect weight changes due to filter handling. For example, if the weight change was large, contamination during transportation or at the sampling site may be occurring; appropriate troubleshooting and corrective actions should be taken.

Data Analysis

The data analyzed were measurements taken from May 2006 to February 2008 for KS1 and from February 2007 to February 2008 for KS2. Data collected when there was a power failure or when instruments had operational problems were discarded. The TEOM values were arithmetically averaged according to the starting and ending time of intensive sampling (e.g., for the 12-h HV/LV sampling duration of 6 a.m. to 6 p.m., the mass concentrations measured by TEOM from 6 a.m. to 6 p.m. were averaged to get the TEOM mass concentration for this run). In some cases, negative values on the TEOM data were observed. These negative values could be due to the nature of particles or instrument malfunction. In accordance with the manufacturer's recommendation, "small" negative values (i.e., 0 to -10 μg m^{-3}), which are likely due to the nature of particles, were considered in the calculation of mean values. "Large" negative values (i.e., <-10 μg m^{-3}) are likely due to instrument malfunction and were discarded in the calculation. Overall, about 14.4% of TEOM raw data had negative values. Of these, about 7.4% (5% for the north or downwind location and 10% for the south or upwind location) had values that are less than <-10 μg/m^3 and 7.0% (6% for the north location and 8% for south location) had values that are within the 0 to -10 μg/m^3. On the basis of the national ambient air quality standards for PM, all measurements were converted to standard conditions of temperature (25°C) and pressure (760 mm of mercury) (U.S.EPA, 2006). Mass concentration data from samplers were compared in pairwise (y vs. x) fashion. In this article, for the comparison of different samplers, concentrations measured by the HV sampler were selected as the reference or the "x" variable (Krieger et al., 2007; Knight and Moore, 1987a, 1987b). The "y" variable represented concentration data from the TEOM or LV samplers. The HV sampler was selected as the reference because it is a FRM sampler. The terms "oversampling" and "undersampling" were then based on the HV sampler as the reference. Comparability was evaluated by linear regression (LR) of y and x with zero intercept. For excellent agreement between samplers, the slope should be close to unity (1±0.1) with a high R^2 value (≥0.94) (Salter and Parsons, 1999; Predicala and Maghirang, 2003; Vega et al., 2003; Chow et al., 2006). Paired t-tests and correlation analyses were also used. The presence of outliers might cause regression errors. Data points with vertical distance from the regression line exceeding four times the standard error (SE) of estimate were eliminated (Cornbleet and Gochman, 1979; Lee et al., 2005). Five and two outliers were eliminated in the comparison of TEOM versus HV samplers at KS1 and KS2, respectively. Three and two outliers were excluded from the comparison of LV versus HV samplers at KS1 and KS2, respectively. The slope (b), R^2 value, and other statistical parameters were generated through Excel (Microsoft Corp, Redmond, Wash.) and SAS (version 9.1.3, SAS Institute Inc., Cary, N.C.).

Results and Discussions

Comparison of Similar PM_{10} Samplers

A total of 60, 41, and 28 data points measured by similar HV, LV, and TEOM samplers were obtained. As shown in table 1, maximum values for the two HV samplers were 395 and 390 μg m^{-3}; minimum values were 9 and 8 μg m^{-3}; and mean values with standard error of mean (SEM) were both equal to 99±13 μg m^{-3}. For the two LV samplers, maximum concentrations were 337 and 302 μg m^{-3}; minimum values were 0 and 13 μg m^{-3}; and average values were 80±11 μg m^{-3} and 77±9 μg m^{-3}. For the TEOM samplers, the average mass concentrations were based on the sampling duration of the HV or LV samplers. Respective maximum values were 145 and 181 μg m^{-3}; minimum values were 6 and 3 μg m^{-3}; and average values were 37±6 μg m^{-3} and 36±8 μg m^{-3}. The t-test of paired samples for means with 95% confidence showed that Pearson correlations were all 0.97 for HV, LV, and TEOM samplers. The P (T<=t) values were greater than 0.05, indicating that similar samplers were highly correlated and there were no significant differences between similar samplers. Figure 2 shows regression lines with slopes of 0.99, 0.90, and 1.05 with R^2 values around 0.98 for HV, 0.97 for LV, and 0.96 for TEOM samplers, respectively.

Table 1. Comparison of similar PM_{10} samplers.

Sampler	n	Min (μg m^{-3})	Max (μg m^{-3})	Mean (±SEM) (μg m^{-3})	R^2	b	SE_b	Paired t-test P-value
HV 1	60	9	395	99(±13)	0.98	0.99	0.02	P>0.05
HV 2		8	390	99(±13)				
LV 1	41	0	337	80(±11)	0.97	0.90	0.02	P>0.05
LV 2		13	302	77(±9)				
TEOM1	28	6	145	37(±6)	0.96	1.05	0.04	P>0.05
TEOM2		3	181	36(±8)				

[a] b is the slope, SE_b is the standard error of b.
[b] SEM is the standard error of the mean.

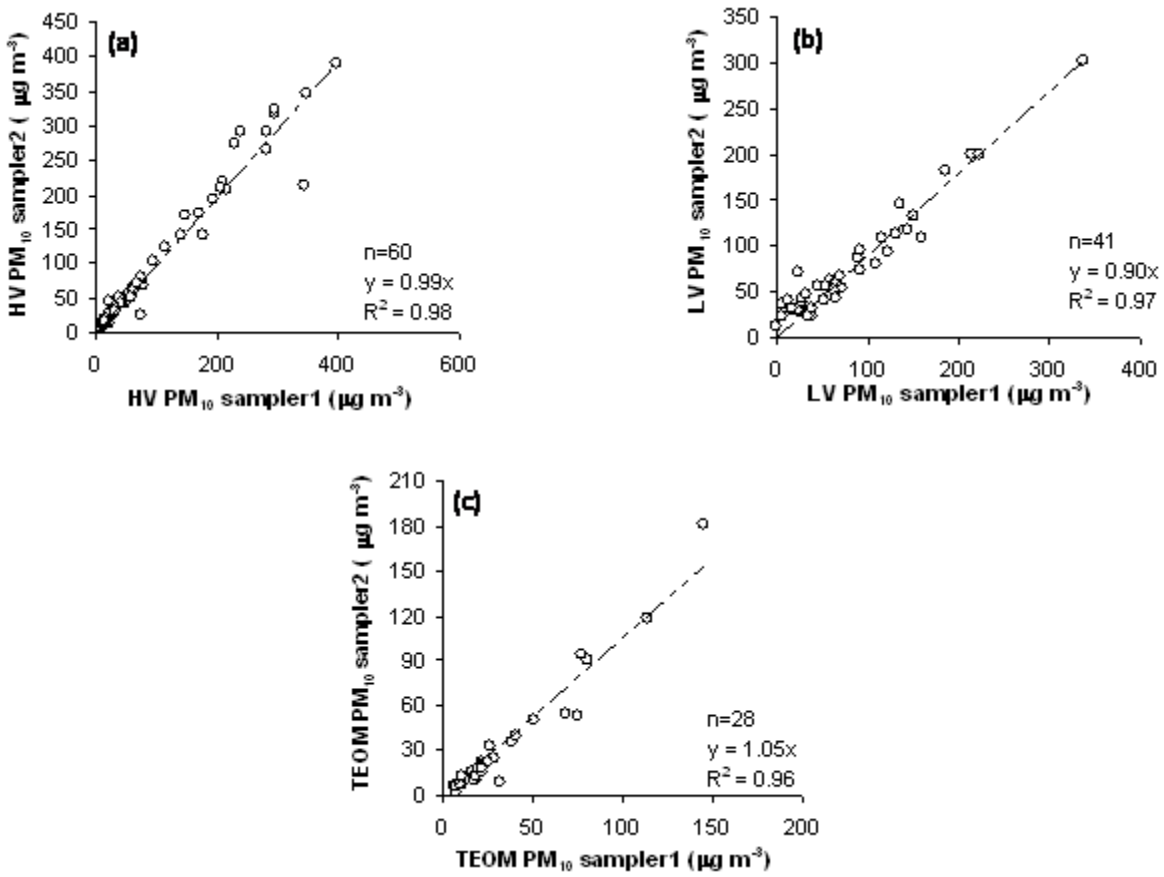


Figure 2. Plots of PM_{10} concentrations for similar samplers at feedlot KS1: (a) high-volume (HV) PM_{10} samplers, (b) low-volume (LV) PM_{10} samplers, and (c) Tapered Element Oscillating Microbalance (TEOM) PM_{10} samplers.

Comparison of TEOM and High-Volume (HV) PM_{10} Samplers

There were 208 and 98 pairs of mass concentration data measured by HV and TEOM samplers at KS1 and KS2, respectively (table 2). At KS1, maximum values of mass concentrations were 537 and 505 μg m^{-3} for the HV and TEOM samplers, respectively; minimum values were less than 10 μg m^{-3}; and corresponding mean values were 76 ± 7 μg m^{-3} and 86 ± 7 μg m^{-3}. At KS2, corresponding maximum values of concentrations were 611 and 713 μg m^{-3}; minimum values were also less than 10 μg m^{-3}; and average values were 47 ± 9 μg m^{-3} and 55 ± 10 μg m^{-3} for HV and TEOM samplers, respectively.

Pearson correlations of 0.96 and 0.90 in t-tests of paired samples indicate that concentrations measured by HV and TEOM samplers at KS1 and KS2 were highly correlated. Slopes of the linear regression (figs. 3a, 3b) were 1.07 and 1.04 with R^2 values of 0.95 and 0.85 for KS1 and KS2, respectively. Comparing KS1 and KS2, the p-value was 0.6 indicating that there was no significant difference in the correlations of mass concentration measured by TEOM and HV samplers between KS1 and KS2. Accordingly, a combined regression line (fig. 3c) was obtained with a slope of 1.06 and R^2 value of 0.93 suggesting that the TEOM tended to oversample when compared with the HV sampler. The p-value obtained by t-test of paired two samples for means was less than 0.05, indicating that there was significant difference between mean concentrations measured by HV and TEOM samplers.

Table 2. Comparison of different PM_{10} samplers.

Sampler	n	Min (μg m^{-3})	Max (μg m^{-3})	Mean (±SEM) (μg m^{-3})	R^2	b	SE_b	Paired t-test P-value
HV-KS1	208	4	537	76(±7)	0.95	1.07	0.02	P < 0.05
TEOM-KS1		5	505	86(±7)				
HV-KS2	98	3	611	47(±9)	0.85	1.04	0.04	P < 0.05
TEOM-KS2		5	713	55(±10)				
HV-KS1	190	2	832	96(±9)	0.88	0.53	0.01	P < 0.05
LV-KS1		0	393	64(±5)				
HV-KS2	63	3	611	68(±13)	0.88	0.59	0.03	P < 0.05
LV-KS2		3	313	52(±7)				

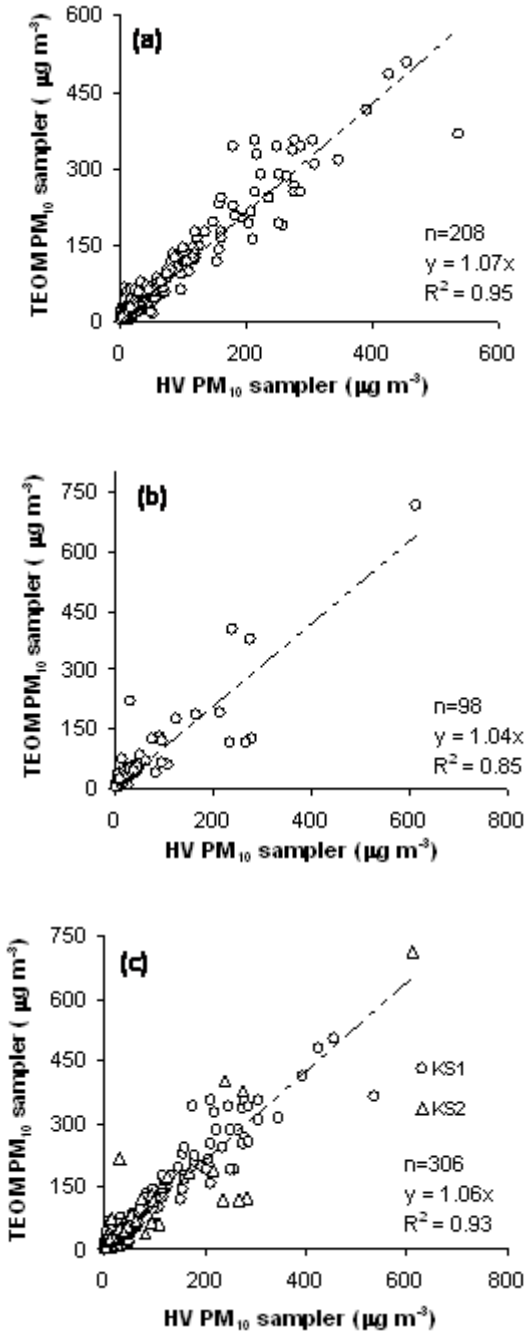


Figure 3. Comparison of *Tapered Element Oscillating Microbalance* (TEOM) and high-volume (HV) PM_{10} samplers for two feedlots: (a) KS1, (b) KS2, and (c) combination of KS1 and KS2.

Since the samplers in this study were placed at two different sampling locations and sampling was conducted in different seasons, better understanding of the performance of these samplers could be made by analyzing the data in groups according to location (i.e., north vs. south) and sampling time (i.e., winter vs. summer). As shown in table 3, for the comparison of TEOM and HV samplers at the south sampling locations of KS1 and KS2, slopes were 1.12 and 1.35 with R^2 values of 0.96 and 0.98, respectively. For the north sampling locations of KS1 and KS2, slopes were 1.04 and 0.99, respectively, with R^2 values of 0.95 and 0.83. These results suggest that TEOM samplers had a greater tendency to oversample at the south sampling location than at the north sampling location.

To determine the effect of season, mass concentration data during winter (December to February) and summer (June to August) were extracted for analysis. Because of limited amount of data, the data from KS1 and KS2 were combined. For the comparison of TEOM and HV samplers, the slope 0.94 with an R^2 value of 0.96 in winter. In summer, the slope was 1.01 with an R^2 value of 0.93 (table 4). This shows that TEOM slightly tended to oversample in summer and undersample in winter compared with the HV sampler.

Effects of meteorological conditions were examined between pairs of measurements at KS1. KS2 was not considered because of limited available weather data taken directly from KS2. During the intensive sampling periods, ambient relative humidity (RH) ranged from 20% to 95% with a mean value of 58.5%; air temperature ranged from -5.9°C to 41.8°C with a mean value of 20.1°C; and wind speed ranged from 0 to 9.0 m s^{-1}, with a mean value of 3.4 m s^{-1}. Figure 4 shows that differences of mass concentration varied only slightly with RH, wind speed, and ambient temperature, with Pearson correlation coefficients of -0.09, 0.20, and 0.09, respectively. There was a slight decrease of concentration difference with RH and an increasing trend of concentration with ambient temperature, as well as wind speed, as shown in figure 4.

The above results indicate that, in general, the TEOM PM_{10} samplers measured higher concentration by approximately 6% than the FRM HV PM_{10} sampler. Under similar conditions, such relationship can be used to correct for sampler bias. Sampler performance is affected by many factors, including sampler characteristics (e.g., sampler inlet design, operating principles), particle characteristics (e.g., shape, size distribution, chemical composition), and weather conditions (e.g., wind speed, temperature). Specific reasons to explain differences in measured concentrations between the TEOM and HV samplers are difficult to ascertain from this study. Previous researchers have observed both undersampling and oversampling by the TEOM compared with the gravimetric method. For example, Vega et al. (2003) measured ambient PM_{10} concentrations from five sites in Mexico City and observed that the TEOM measured larger PM_{10} concentrations than the gravimetric method. They attributed the larger TEOM PM_{10} concentration to

Table 3. Comparison of different PM_{10} samplers for different locations.

		TEOM vs. HV				LV vs. HV			
		n	R^2	b	SE_b	n	R^2	b	SE_b
KS1	N	102	0.95	1.04	0.02	58	0.96	0.50	0.01
	S	106	0.96	1.12	0.02	58	0.93	0.61	0.02
KS2	N	44	0.83	0.99	0.07	31	0.91	0.53	0.03
	S	54	0.98	1.35	0.03	32	0.95	0.85	0.04

Table 4. Comparison of different PM_{10} samplers for different seasons.

		n	R^2	b	SE_b
Summer	TEOM vs. HV	117	0.93	1.01	0.03
Winter		70	0.96	0.94	0.02
Summer	LV vs. HV	34	0.94	0.49	0.02
Winter		49	0.77	0.90	0.07

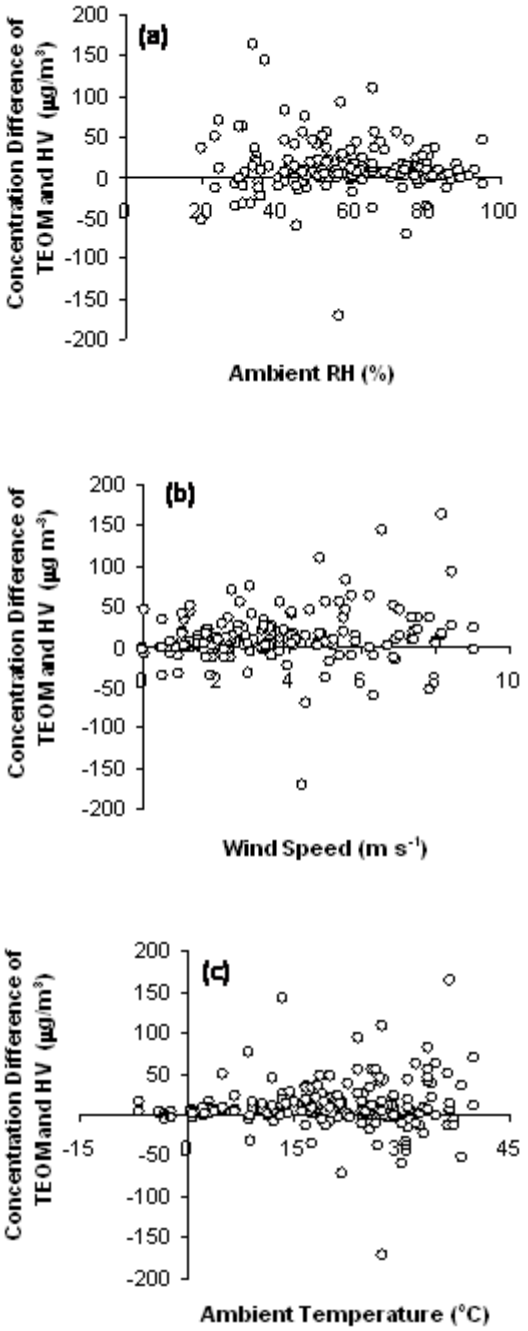


Figure 4. Effects of weather conditions on the difference of *Tapered Element Oscillating Microbalance* (TEOM) and high-volume (HV) PM_{10} mass concentrations: (a) ambient relative humidity (RH), (b) wind speed, and (c) ambient temperature.

re-entrainment of larger particles from the unclean TEOM PM_{10} sampler inlet. In this study, routine cleaning of the TEOM PM_{10} inlet was conducted during the sampling period; thus, the possibility of re-entrainment was reduced. Other researchers (e.g., Bulpitt and Price, 2006) have observed undersampling of the TEOM and attributed it to the loss of semi-volatile organic compounds and particle-bound water during sampling.

Chow et al. (2006) noted that particulate monitors have higher comparability when particles are chemically stable, small in size, and when the particles having diameters similar to the sampling inlet cut point are not dominant. Other researchers (Wang et al., 2005; Wanjura, 2005; Buser et al., 2007) also reported that changes in particle size distribution and concentration of PM_{10} resulted in various oversampling and undersampling errors. In this study, the particle size distribution at the downwind sampling location was measured with two cascade impactors; the mean mass median diameter (MMD) was 12 µm, which was larger than the 10-µm cut point of the size-selective inlet. The larger MMD might have caused the TEOM PM_{10} sampler to measure higher values (Buser et al., 2007) than the HV sampler. Note that collection efficiency of the PM_{10} samplers may be affected by the inlet cut point and velocity of impaction (Shaw et al., 1983; McFarland et al., 1984). O'Shaughnessy et al. (2007) suggested that particle shape, density, and other physical factors have a greater effect on sampler bias than environmental factors.

Comparison of Low-Volume (LV) and High-Volume (HV) PM_{10} Samplers

As shown in table 2, there were 190 and 63 pairs of mass concentrations measured by HV and LV samplers at KS1 and KS2, respectively. For KS1, maximum mass concentrations measured by HV and LV were 832 and 393 µg m^{-3}; minimum values were less than 10 µg m^{-3}; and average values were 96 ± 9 µg m^{-3} and 64±5 µg m^{-3}, respectively. For KS2, maximum values were 611 and 313 µg m^{-3}; minimum values were less than 10 µg m^{-3}; and average values were 68±13 µg m^{-3} and 52±7 µg m^{-3}.

Concentrations measured by LV and HV samplers were highly correlated with Pearson correlations of 0.91 and 0.92 in the paired t-test for KS1 and KS2, respectively. Correlation was improved for sampling durations of 12 h or longer at KS1, which had a Pearson correlation of 0.97. Longer sampling duration seemed to improve correlation of the LV with the HV sampler. This was one reason why the sampling duration was increased from 2 to 6 h to at least 12 h since August 2007. Figure 5 summarizes the relationship between the LV and HV samplers. A significant difference was observed with p-values less than 0.05 and low slopes of about 0.5 in the regression lines. Analysis also indicated that there was no significant difference in the correlation of mass concentration measured by LV and HV samplers between KS1 and KS2 ($Pr > 0.05$). Combined data measured by LV and HV samplers at KS1 and KS2 had a linear function with slope of 0.53 and R^2 value of 0.93 (fig. 5c). Unlike the TEOM, which seemed to have the tendency to oversample compared with the HV sampler, the LV samplers tended to significantly undersample compared with the HV samplers.

The effects of sampling location and season on the performance of the LV and HV samplers are summarized in tables 3 and 4, respectively. Regardless of sampling location and season, the LV samplers tended to undersample compared with the HV samplers. The difference in

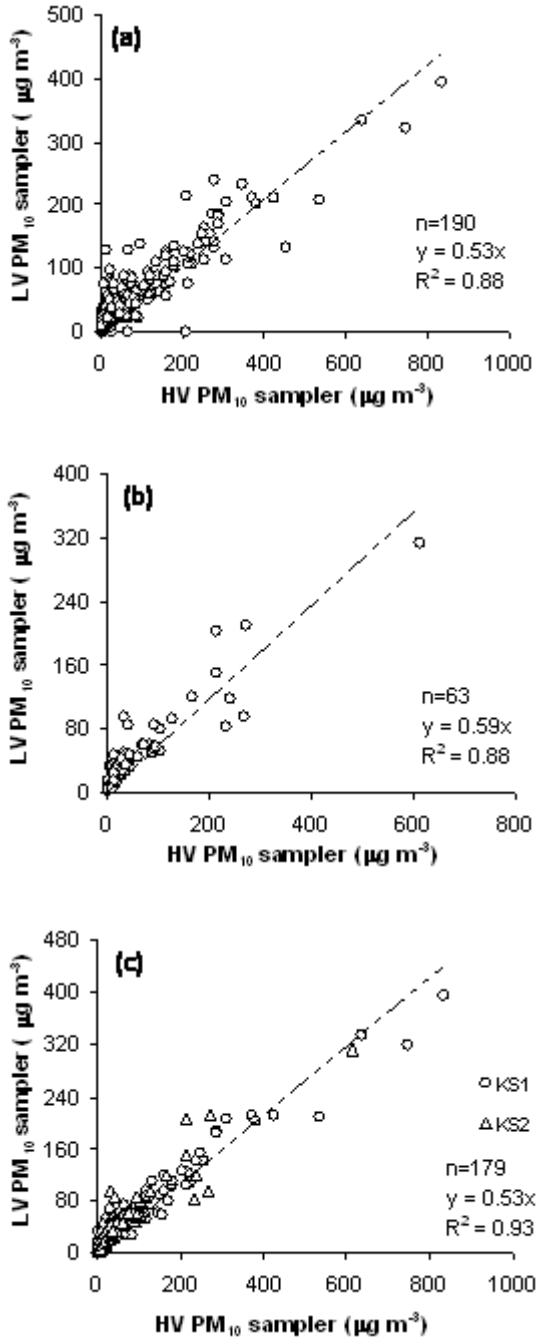


Figure 5. Comparison of high-volume (HV) and low-volume (LV) PM_{10} samplers: (a) for feedlot KS1, (b) for feedlot KS2, and (c) combination of KS1 and KS2 with sampling duration ≥12 h.

performance between the LV and HV samplers were affected by sampling location and season. For example, the degree of undersampling of the LV sampler compared with the HV sampler was greater in summer than in winter as indicated by the smaller regression coefficient in summer than in winter (0.49 vs. 0.90). Also, the degree of undersampling was greater for the north sampling location than for the south sampling location. Similar to the TEOM-HV sampler comparison, the difference in concentration between the LV and HV samplers were only slightly affected by weather conditions, including RH, wind speed, and temperature.

In general, the LV PM_{10} sampler showed much smaller concentration (by about 47%) compared with the HV PM_{10} sampler. Under similar conditions, such relationship could be used to account for sampler bias. Similar to the comparison between the TEOM and HV samplers, it is likely that the results were due to the interaction between inherent sampler characteristics and particle size distribution (Wang et al, 2005; Buser et al., 2007). It should be noted that the LV samplers were likely more prone to experimental errors than the HV samplers because of the small amounts of PM_{10} that were collected on the filters as a result of lower sampling flow rate (5 L min^{-1}). While there appears to have no guidelines on the lower concentration limit for PM_{10}, there is a guideline for $PM_{2.5}$ (U.S. EPA, 1998). Based on the guideline, the estimated amount of dust that must be collected on the filters is about 48 µg. In this study, approximately 86% out of 154 data points for the LV samplers collected on the north (downwind) location and 74% out of 152 data points collected on the south (upwind) location had mass values ≥ 48 µg.

It should also be noted that the HV PM_{10} sampler was used as the reference, primarily because it was a FRM sampler. The HV PM_{10} sampler has sampler bias in itself, as documented by several researchers (e.g., Buser et al., 2007), particularly if the particles being sampled are large. Clearly, there is a need to further evaluate the performance of the samplers considered in this study as affected by particle characteristics and operating conditions.

Conclusion

This research compared three types of PM_{10} samplers. The following conclusions were drawn:

- Collocated PM_{10} samplers showed similar trends but significant differences in concentrations.
- The TEOM PM_{10} sampler (a federal equivalent method sampler) had great correlation with the federal reference method high-volume PM_{10} sampler. In general, the TEOM tended to oversample PM_{10} (by approximately 6%) compared with the high-volume sampler.
- The low-volume PM_{10} sampler tended to undersample (by about 47%) compared with the high-volume sampler.
- Differences in PM_{10} concentration among samplers slightly decreased with an increase in relative humidity and decreased with an increase in wind speed and ambient temperature.

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